

CANDY FACTORY

CHAPTER 3: FACTOR FRIENDS

FACTORS • NUMBER SQUAD • HCF • LCM

"Master the building blocks of numbers!"

CANDY - 1

Factors & Multiples

HOW TO SOLVE:

1. Start from 1. Divide the number.
2. If remainder is 0, it is a Factor.
3. Write them in pairs (Ex: 1x12, 2x6).
4. Multiples are found by multiplying the number by 1, 2, 3...

Avoiding Mistakes:

- Missing factors in the middle of a list.
- Confusing Factors (small) with Multiples (large).
- Forgetting 1 and the number itself.

Example: Factors of 24

Step 1: Start with 1: 1 x 24.

Step 2: Check 2: 2 x 12.

Step 3: Check 3: 3 x 8.

Step 4: Check 4: 4 x 6.

Step 5: List all numbers from the pairs.

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Pairs: 1x24, 2x12, 3x8, 4x6
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Final Result: 1, 2, 3, 4, 6, 8, 12, 24

Example: Factors of 18

Step 1: 1 x 18.

Step 2: 2 x 9.

Step 3: 3 x 6.

Step 4: List in order.

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Pairs: 1x18, 2x9, 3x6
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Final Result: 1, 2, 3, 6, 9, 18

Example: Multiples of 15

Step 1: 15 x 1 = 15.

Step 2: 15 x 2 = 30.

Step 3: 15 x 3 = 45.

Step 4: Keep going!

Final Result: 15, 30, 45, 60, 75

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15, 30, 45, 60, 75...
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Example: Multiples of 20

Step 1: 20 x 1, 20 x 2...

Step 2: Multiples never end!

Step 3: List the first 5.

Final Result: 20, 40, 60, 80, 100

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20, 40, 60, 80, 100...
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EXERCISE 1: STANDARD SUMS

1. Factors of 16
2. Factors of 45
3. Factors of 50
4. Factors of 64
5. Factors of 81
6. Multiples of 7 (First 5)
7. Multiples of 13 (First 5)
8. Is 5 a factor of 125?
9. Common factors of 12 & 18
10. Multiples of 9 between 20-50
11. Factors of 100
12. Factors of 21
13. Factors of 36
14. Multiples of 25 (First 4)
15. Is 8 a factor of 72?
16. Common multiples of 4 & 6
17. Factors of 12
18. Factors of 28
19. Multiples of 11 (First 5)
20. Factors of 75
21. Factors of 90
22. Is 10 a multiple of 5?
23. Smallest factor of 500
24. Largest factor of 42
25. How many factors has 17?

EXERCISE 2: STORY PROBLEMS

1. A box has 24 chocolates. How many equal ways to group?
2. A bus comes every 10 mins. List times in 1 hour.
3. Flower shop has 36 roses. Can we make 5 equal bunches?
4. Pencils come in packs of 12. Buy 8 packs, total?
5. Find a number between 10-20 that is a multiple of 3 & 6.
6. Teacher has 48 books for 6 students. Equal sharing?
7. Stickers come in multiples of 5. Can I buy 23 stickers?
8. Identify common factors of my age (10) and 20.
9. Bell rings every 15 mins. How many times in 2 hours?
10. Which has more factors: 12 or 15?

Tips & Tricks:

- Every number has 1 as its smallest factor.
- Even numbers always have 2 as a factor.
- Multiples are infinite like the sky!
- 1 is the only number with exactly one factor.

CANDY - 2

The Number Squad

IDENTITY RULES:

Even: Ends in 0, 2, 4, 6, 8. Divisible by 2.

Odd: Ends in 1, 3, 5, 7, 9. Not divisible by 2.

Prime: Has ONLY 2 factors (1 and itself).

Composite: Has MORE than 2 factors.

Twin Primes: Two primes with a gap of 2 (Ex: 3, 5).

Co-Primes: Numbers sharing ONLY 1 as common factor.

Avoiding Mistakes:

- Thinking 1 is Prime. (1 is neither Prime nor Composite!)
- Thinking all Odd numbers are Prime. (Look at 9 or 15!)
- Thinking Co-primes must be prime numbers.

Example: Is 13 Prime or Composite?

Step 1: List factors of 13.

Step 2: Pairs: 1 x 13.

Step 3: Only 2 factors found.

Step 4: Conclusion: It is Prime!

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Factors: 1, 13
```

Final Result: Prime

Example: Are 8 and 15 Co-prime?

Step 1: Factors of 8: 1, 2, 4, 8.

Step 2: Factors of 15: 1, 3, 5, 15.

Step 3: Common factor: Only 1.

Step 4: Result: Co-prime!

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8: {1, 2, 4, 8} | 15: {1, 3, 5, 15}
```

Final Result: Yes

Example: Is 21 Composite?

Step 1: Factors: 1, 3, 7, 21.

Step 2: Total: 4 factors.

Step 3: Since > 2, it is Composite.

Final Result: Yes

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Factors: 1, 3, 7, 21
```

Example: Twin Prime Check (11, 13)

Step 1: Check if 11 is Prime (Yes).

Step 2: Check if 13 is Prime (Yes).

Step 3: Gap: 13 - 11 = 2.

Step 4: Result: Twin Primes!

Final Result: Yes

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11 & 13 (Gap is 2)
```

EXERCISE 1: IDENTITY QUIZ

1. Is 2 Prime or Composite?
2. List first 5 Prime numbers
3. List Odd numbers between 20-30
4. Is 51 Prime? (Check 3x17!)
5. Are 14 and 15 Co-prime?
6. Smallest Composite number
7. Only Even Prime number
8. List 3 pairs of Twin Primes
9. Is 0 Even or Odd?
10. Composite numbers below 10
11. Next Prime after 19
12. Is 27 Composite?
13. Prime numbers between 1-10
14. Are 4 and 9 Co-prime?
15. Largest 2-digit Prime number
16. Is 91 Composite? (Check 7x13!)
17. Sum of first 3 Odd numbers
18. Twin Primes between 10-20
19. Next Even after 98
20. Is 1 a Prime number?
21. Smallest Odd Composite number
22. Are 21 and 25 Co-prime?
23. Factors of 2 Prime numbers
24. Next Composite after 15
25. Are 12 and 13 Co-prime?

EXERCISE 2: IDENTITY STORIES

1. I am a Prime between 10-20. Digits add to 2. Who?
2. Class of 23 kids. Can we group into 4 equal teams?
3. Smallest Odd Composite number. Who am I?
4. List all Twin Primes below 30.
5. Is my house number 45 Prime or Composite?
6. Add Odd + Odd. Result is?
7. Find two Composite numbers that are Co-prime.
8. Why is 2 the 'Hero' of Prime numbers?
9. Basket of 15 apples. Can 4 friends share exactly?
10. Name a number that is neither Prime nor Composite.

Tips & Tricks:

- 2 is the only Even Prime number.
- All Twin Primes except (3, 5) have an even number between them.
- Any two consecutive numbers are always Co-prime.
- Composite numbers have factors other than 1 and self.

CANDY - 3

HCF - The Great Divider

LADDER METHOD (L-DIVISION):

Write numbers side by side.

Divide both by a COMMON prime (2, 3, 5...).

Stop when no more common divisors exist.

Multiply ONLY the divisors on the LEFT side.

Avoiding Mistakes:

- Multiplying the bottom remainder numbers (That's for LCM!).
- Stopping too early when a common factor still exists.
- Not using Primes to divide.

Example: HCF of 24 and 36

Step 1: Commonly divide by 2: (12, 18).

Step 2: Commonly divide by 2 again: (6, 9).

Step 3: Commonly divide by 3: (2, 3).

Step 4: Multiply Left Side: 2 x 2 x 3 = 12.

Final Result: 12

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2 | 24, 36
2 | 12, 18
3 | 6, 9
  | 2, 3
```

Example: HCF of 15 and 25

Step 1: No common factor 2 or 3. Use 5.

Step 2: Divide by 5: (3, 5).

Step 3: Stop. HCF = 5.

Final Result: 5

```
5 | 15, 25
  | 3, 5
```

Example: HCF of 12 and 16

Step 1: Divide by 2: (6, 8).

Step 2: Divide by 2: (3, 4).

Step 3: Stop. HCF = 2 x 2 = 4.

Final Result: 4

```
2 | 12, 16
2 | 6, 8
  | 3, 4
```

Example: HCF of 10 and 20

Step 1: Divide by 2: (5, 10).

Step 2: Divide by 5: (1, 2).

Step 3: HCF = 2 x 5 = 10.

Final Result: 10

```
2 | 10, 20
5 | 5, 10
  | 1, 2
```

EXERCISE 1: HCF HUNT

1. HCF of 12, 18
2. HCF of 20, 30
3. HCF of 15, 45
4. HCF of 40, 60
5. HCF of 14, 21
6. HCF of 8, 20
7. HCF of 36, 54
8. HCF of 10, 25
9. HCF of 16, 24
10. HCF of 22, 33
11. HCF of 50, 75
12. HCF of 13, 26
13. HCF of 9, 27
14. HCF of 4, 8, 12
15. HCF of 10, 20, 30
16. HCF of 18, 24
17. HCF of 7, 14
18. HCF of 25, 100
19. HCF of 35, 49
20. HCF of 11, 22
21. HCF of 48, 72
22. HCF of 6, 12, 18
23. HCF of 21, 28
24. HCF of 100, 200
25. HCF of 15, 20

EXERCISE 2: STORY PROBLEMS

1. Two ropes 12m and 18m. Max equal pieces?
2. 30 boys, 45 girls in teams. Max team size?
3. Fill 20L and 30L cans with same cup. Max size?
4. Tiles for 12ft by 16ft floor. Max square tile?
5. 60 apples, 80 oranges. Max fruits per bag?
6. divide ₹100 and ₹150 equally. Max amount?
7. String A: 24cm, B: 36cm. Max equal cuts?
8. Box A: 15kg, B: 25kg. Max common weight?
9. Groups of 12 and 18. Max kids per row?
10. Smallest divisor for 13 and 17.

Tips & Tricks:

- HCF of Co-primes is always 1.
- HCF is also called GCD (Greatest Common Divisor).
- HCF is always <= the smallest number.
- HCF divides all given numbers perfectly.

CANDY - 4

LCM - The Meeting Point

COMMON DIVISION LOGIC:

Divide by a prime that fits AT LEAST ONE number.

Bring down any number that isn't divisible.

Continue until all numbers become 1.

Multiply ALL numbers on the LEFT side.

Avoiding Mistakes:

- Stopping too early (Wait until all are 1!).
- Forgetting to multiply all the factors.
- Bringing down a number and dividing it by mistake.

Example: LCM of 12 and 15

Step 1: Divide by 2: (6, 15). 15 brought down.

Step 2: Divide by 2: (3, 15).

Step 3: Divide by 3: (1, 5).

Step 4: Divide by 5: (1, 1).

Step 5: LCM = 2x2x3x5 = 60.

Final Result: 60

```
2 | 12, 15
2 | 6, 15
3 | 3, 15
5 | 1, 5
  | 1, 1
```

Example: LCM of 4 and 6

Step 1: Divide by 2: (2, 3).

Step 2: Divide by 2: (1, 3).

Step 3: Divide by 3: (1, 1).

Step 4: LCM = 2x2x3 = 12.

Final Result: 12

```
2 | 4, 6
2 | 2, 3
3 | 1, 3
  | 1, 1
```

Example: LCM of 10 and 15

Step 1: Divide by 2: (5, 15).

Step 2: Divide by 3: (5, 5).

Step 3: Divide by 5: (1, 1).

Step 4: LCM = 2x3x5 = 30.

Final Result: 30

```
2 | 10, 15
3 | 5, 15
5 | 5, 5
  | 1, 1
```

Example: LCM of 8 and 12

Step 1: Divide by 2: (4, 6).

Step 2: Divide by 2: (2, 3).

Step 3: Divide by 2: (1, 3).

Step 4: Divide by 3: (1, 1).

Step 5: LCM = 24.

Final Result: 24

```
2 | 8, 12
2 | 4, 6
2 | 2, 3
3 | 1, 3
  | 1, 1
```

EXERCISE 1: LCM RACE

1. LCM of 4, 6
2. LCM of 8, 12
3. LCM of 10, 15
4. LCM of 12, 18
5. LCM of 5, 7
6. LCM of 20, 25
7. LCM of 6, 9
8. LCM of 14, 21
9. LCM of 16, 24
10. LCM of 3, 4, 5
11. LCM of 11, 13
12. LCM of 2, 8, 10
13. LCM of 15, 20
14. LCM of 30, 40
15. LCM of 9, 12
16. LCM of 7, 14
17. LCM of 10, 20, 30
18. LCM of 25, 50
19. LCM of 12, 16
20. LCM of 5, 10, 15
21. LCM of 18, 27
22. LCM of 8, 10
23. LCM of 40, 60
24. LCM of 2, 3, 4
25. LCM of 100, 150

EXERCISE 2: STORY PROBLEMS

1. Two bells ring every 4 and 6 mins. When together?
2. Bus A: 10 mins, Bus B: 15 mins. Next meet?
3. Light A flashes every 8s, B every 12s. Common flash?
4. Steps: 40cm and 50cm. When do they align?
5. Smallest number divisible by 12 and 15.
6. Hotdogs (8) and Buns (10). Min to match?
7. Frog jumps 3, Toad jumps 5. Common landing?
8. Meeting at start: 3 and 4 mins per lap.
9. Smallest number with 0 remainder for 6 and 9.
10. Signals turn green every 20s and 30s. Next time?

Tips & Tricks:

- LCM is used for things that happen 'again and again'.
- HCF x LCM = Product of two numbers.
- LCM of Co-primes is just their Multiplication.
- LCM is always >= the largest number.

MIXED MARATHON!

25 FINAL MASTER CHALLENGES

MASTER LEVEL CHALLENGES

1. HCF of 24 and 32.
2. LCM of 6 and 8.
3. Is 105 Prime or Composite?
4. Smallest 3-digit multiple of 12.
5. Are 18 and 35 Co-prime?
6. List factors of 36 that are Even.
7. HCF of 100 and 101.
8. LCM of 10, 100, 1000.
9. Smallest number divisible by 2, 3, 5.
10. Product of numbers is 48. HCF is 2. LCM?
11. Is 231 divisible by 3?
12. Sum of all factors of 6.
13. Twin Primes between 1 and 20.
14. Is every multiple of 4 also a multiple of 2?
15. HCF of two Co-primes is what?
16. LCM of 11 and 13.
17. Factors of 64.
18. Multiples of 15 between 50-100.
19. Greatest factor of 80.
20. Smallest multiple of 99.
21. Are 8 and 21 Co-prime?
22. HCF of 15, 30, 45.
23. Sum of first 5 Prime numbers.
24. Is 99 Prime or Composite?
25. LCM of 12 and 16.

THINK & TRY

EASY

List 3 numbers with only 2 factors. Name them.

MEDIUM

HCF is 1, LCM is 35. What are the numbers?

HARD

2-digit multiple of 6. Digit sum is 9. Factors are 9 & 18. Who?